

Who reports the phish?

A critique of Petrič & Just (2025): information security culture and employees' reporting of phishing emails

Verdict: theoretically coherent and statistically sophisticated, but a cross-sectional, self-report design limits causal and behavioural claims.



employee spots → reports → SOC contains

The study at a glance

From “how do we stop clicks?” to “how do we get employees to report?”



Problem

Phishing susceptibility is well researched; the drivers of reporting are not.



Objectives

Test whether four ISC components predict reporting via attitudes and responsibility, and whether the model holds in DE, UK and US.



Significance

Timely reports let security teams contain attacks before financial loss and disruption.

1,017

employees surveyed

3

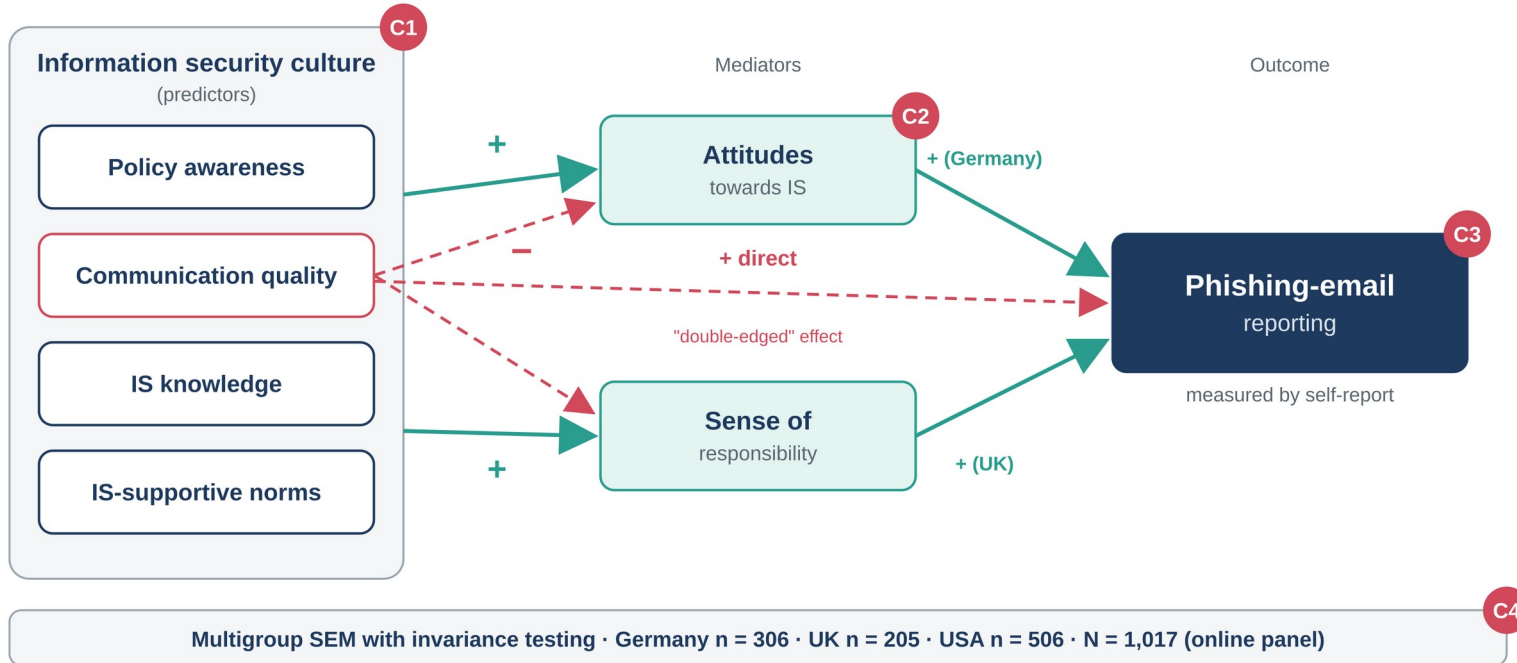
countries: DE • UK • US

4

culture components tested

What they tested, and where it is vulnerable

Multigroup SEM; red badges mark the four methodological concerns



Key finding: norms, policy awareness and knowledge raise attitudes and responsibility, which raise reporting. Communication quality is “double-edged”: it lifts reporting directly but lowers responsibility and attitudes.

Methodology scorecard

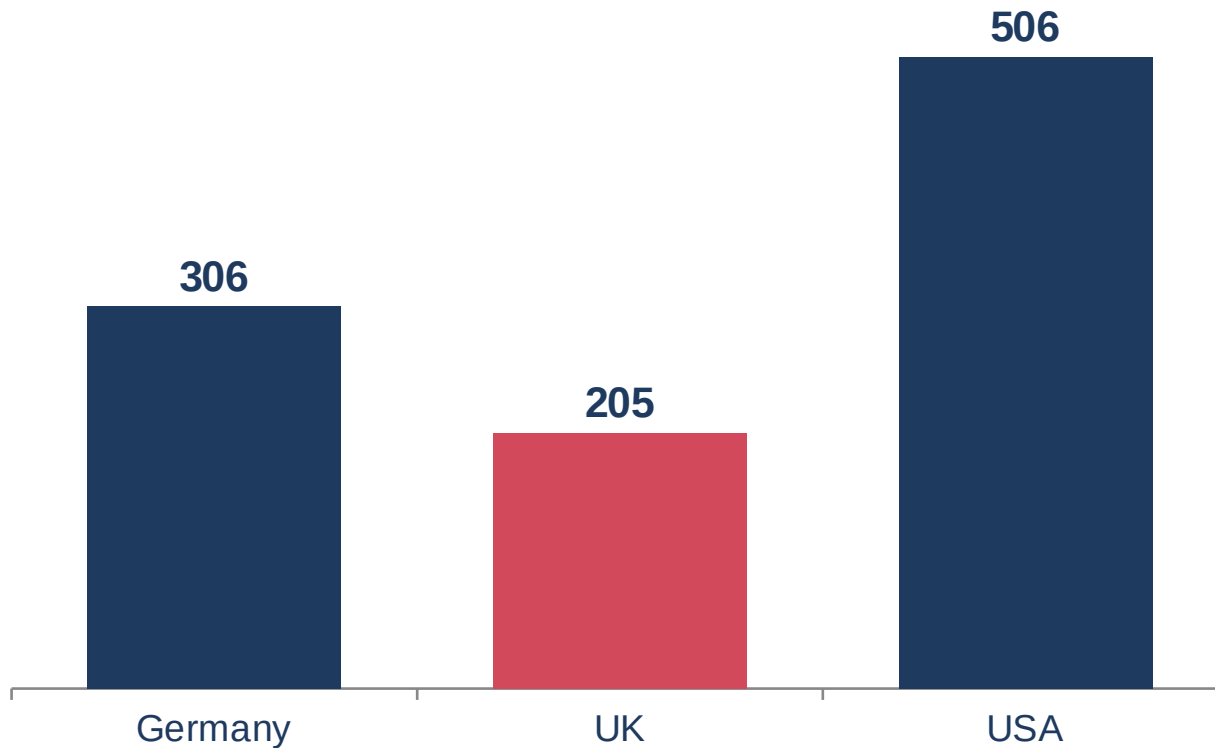
Each element judged against methods literature

Element	What they did	Verdict	Why it matters
 Design	Deductive, cross-sectional survey	Fit, but limited	Suits theory testing, but mediation needs time order (Maxwell & Cole, 2007)
 Sampling	Online panel, orgs ≥ 50 staff; $n = 306 / 205 / 506$	Weak	Non-probability, self-selected; unequal groups cut power (Callegaro et al., 2014)
 Data collection	One self-report questionnaire for all constructs	Weak	Common method & social desirability bias (Podsakoff et al., 2003)
 Statistics	Multigroup SEM with invariance testing	Strong	Invariance is a prerequisite for valid cross-country comparison (Vandenberg & Lance, 2000)

Sampling: who was actually asked?

Large overall, but uneven and non-probability

Respondents per country (N = 1,017)



UK group is 60% smaller than the US group

Less precise UK paths and less power to detect country differences.



Online panel members self-select

May be more digitally literate and security-aware than typical staff.



One voice per organisation (apparently)

An organisational culture is inferred from individual perception.

Strengths vs limitations

Judged on validity, reliability and generalisability



Strengths

- Theory-driven, parsimonious model with manageable levers
- Large three-country sample with demonstrated invariance
- Indirect paths reveal the “double-edged” communication effect
- Authors are candid about the weak outcome measure

Limitations

Internal validity

No temporal order: reporting may shape attitudes, not vice versa (Shadish et al., 2002)

Construct validity

Self-reported reporting ≠ behaviour: the intention–behaviour gap (Sheeran, 2002)

Reliability

One occasion only; shared-method variance may inflate estimates

Statistical conclusion

$N > 1,000$ makes tiny paths significant; report effect sizes

Generalisability

Three Western economies, larger firms; country ≠ culture

Four evidence-based improvements

Each targets a specific concern



Measure real behaviour

C3

Link surveys to simulated-phishing report logs (Lain et al., 2022)



Separate the time points

C2

Culture at T1 → attitudes at T2 → reporting at T3 (Maxwell & Cole, 2007)



Go multilevel

C1

Several employees per organisation; multilevel SEM; power-planned strata (Faul et al., 2007)



Measure culture, widen reach

C4

Measure cultural values directly; add non-Western and smaller firms

Trade-off: *stronger causal and behavioural evidence costs more and raises ethical questions about deception in phishing simulations.*

So what for business research?



Valuable contribution

Reframes employees as active defenders and tests the model rigorously across three countries.



Association, not causation

A cross-sectional, self-report design shows links between perceptions, not what drives behaviour.



Next step

Multilevel, longitudinal and behavioural designs; managers should pilot the levers as testable hypotheses.

Thank you. Questions?

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